American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

AIUB UNIFIED RESOURCES APPLICATION (AURA)

Semester: Fall 25-26

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1. PROJECT PROPOSAL

1.1 Background to the Problem

At AIUB, it is a very regular phenomenon and basic needs to use AIUB portal. Almost every work is done by this platform. But it is noticed that, students often face difficulties to log in portal especially when the server is overloaded in high traffic and portal seems very slow. And it can be noted that these difficulties greatly seen in student's registration time. On the addition, Faculty members also feel difficulties to publish result and importance notice because of server delay. Sometimes the administrators also face challenges to do their relative task or management. And the most important thing that, there is no online process to boking a slot on playground ,swimming pull and indoors too.so it forced one in person to visit office for booking a desire slot which is not only time consuming but also inconvenient.

To address this issue *AIUB UNIFIED RESOURCES APPLICATION* **(AURA)** has been proposed — an Android-based application designed to integrate all academic, administrative, and co-curricular services into a single, unified platform.

The app aims to reduce server load, provide faster access by doing information up-to-date and ensure better performance. It also includes role-based access, allowing students, teachers, supervisors, and administrators to perform their respective tasks efficiently. By solving problems like routine checking, server overload, and slot booking difficulties, *AURA*

also offers online slot booking system. Ultimately, this project provides a reliable, user-friendly, and efficient system that enhances communication and overall management across the university.

1.2 Selection of Process Model

We have selected **Scrum** for our project as it meets our needs.

The project is developed for AIUB where students, faculties and admins use the portal daily for academic and administrative tasks. It includes different roles (student, faculty, admin, supervisor) with different needs. The environment is dynamic because university activities, schedules and rules change. The requirements include routine checking, slot booking, result publishing, profile creation and communication features. These are **functional requirements** that must be reliable and easy to use. **Non-functional requirements** include speed, server load management and security. Some requirements are stable such as login, routine checking, result publishing and profile management. But some are likely to change like slot booking rules, class representative election process or new admin activities. Because universities update policies and add new services. project has both stable and changing requirements. That's why using an agile method like **Scrum** is suitable as it allows quick adaptation to new need.

We have selected **Scrum** because it supports our small team size of 4 members. Scrum uses **Sprints** where tasks are divided into smaller parts and assigned clearly which helps us to manage work easily. Daily meetings keep communication open so everyone knows what others are doing. The roles and backlog in Scrum make co-ordination simple and tasks can be adjusted if requirements change. This model is feasible to meet the business objective because it allows us to deliver working features step by step, test them early and improve based on feedback. In the end, Scrum ensures teamwork, faster development and a user-friendly solution for the university portal.

Scrum is very flexible in adapting to changes. Since work is done in short sprints, new features or changes in scope can be added in the next sprint without disturbing the whole project. If technology updates or better tools appear, they can be integrated during sprint planning. User requirements can also be adjusted easily because Scrum takes regular feedback and updates the backlog. This flexibility makes it suitable for a dynamic project like a university portal, where rules, services, or user needs may change.

2. SOFTWARE REQUIREMENTS SPECIFICATIONS (SRS) / PRODUCT REQUIREMENTS DOCUMENT (PRD)

2.1 Scopes and Features

Scope 1:

The project will focus on creating an Android-based portal for students, faculties, supervisors and admins.

Scope 2:

It will provide academic, administrative, and co-curricular services.

Scope 3:

The app will reduce server load by storing data locally.

Scope 4:

It will not have money or payment options.

Scope 5:

The app will be used only for AIUB.

Project Features:

1. Login System
2. Dashboard
3. Profile Creation and Management
4. Course and Section Management
5. Routine Viewing
6. Drop Application Submission and Review
7. Playground Slot Booking
8. Club Joining and Member Review
9. Class Representative Election Participation
10. Push notifications
11. Result publishing
12. Approving slot booking
13. Communication with OSA

2.2 User Story Table

As A/An	Want to	So that	Acceptance Criteria
Admin	Create faculty profile	I can maintain faculty information.	Valid name, email and faculty details.
	Creation sections	I can assign faculties to their respective sections.	Valid section
	Manage sections	I can change any mistakes that has made during creation.	Valid sections
Student Admin	Create student profile	I can maintain students' information.	Valid name, email, student ID and other information's.
	Update student course	I can help them regarding their course issues.	Valid student with issue regarding course.
Co-Curricular Admin	Review requested slots for playgrounds	I can schedule playground time.	Valid slot booking.
Supervisor	Create admin profile	I can maintain admin information's.	Valid name, email and other information's.
	Create/modify course	I can add and remove courses.	Valid course name and Course ID.
Faculty	Review my section's students	I can maintain their curriculum.	Valid section with valid students.
	Review drops applications	I can approve or deny student's course drop application.	Valid drop application.
	Push notifications	I can keep updated my students about upcoming curriculum activities like quizzes or important dates.	Valid section with valid students.
	Push results	I can keep track of a student's progress.	Valid student within the section.
Class Representative	Push notifications	I can push important notifications on behalf of respected faculty.	Valid student within the section and already elected as CR.
Club Representative	Review club members	I can review my club members who has joined already	Valid student within same club.

Student	Access class routine	I can attend classes without any hassle	Valid student
	Join clubs	I can do my extracurricular activities	Valid student, valid club
	Book playground slot	I can practice outdoor games	Valid student, valid slot selection.
	Attend class representative election	I can help my section	Valid student, valid section
	Communicate student. officials	I can seek for help anytime I face problem	Valid problem

2.3 Requirements Traceability Matrix

AURA will make the University System simple for the students, faculties and authorities as well. It will also reduce the server side pressure as it has offline issues. Generating routine for students is one of the major features of AURA that will solve the problem of time-based session out and server pressure. Students can also book slots and join clubs more easily than before. It will have a class representative who will be selected by an election among the students of the section. The class representative will help respected faculty to push notifications based on important dates and events.

Each workflow is designed to be simple, quick and user-friendly, helping users complete their tasks smoothly without any confusion. For Faculty admins, they can create faculty profile. They can create/modify/delete sections and assign faculties to those sections. If a faculty member wants to publish results, they have to log into their faculty account, open their assigned section, upload the grades and send notifications to students. If a student wants to book a sports slot, they have to log into the app as their student id, go to the co-curriculum section, choose a valid available time and submit a request for to approve. The co-curricular admin they review slot requests that requested by student.

2.3.1 Functional Requirements

Offline Routine View: Students can check their routine in offline through this app.
QR Club Joining: Students can easily join to any group by scanning QR code.
Slot Booking: Students can book for available slot by this app.
Notification System: All users can get instant update, notifications.
Class Representative: They can push important notifications to their section on-behalf of their respective faculties.
Online Slot Booking: Students can book playground slot from the app easily.

Representation requirements using user stories or use case style descriptions: ~

Faculty Admin Use Cases:

- As A Faculty Admin I can maintain faculty information.
- As A Faculty Admin I can assign faculties to their respective sections.
- As A Faculty Admin I can correct any mistakes that occurred during profile creation.

Faculty Use Cases:

- As A Faculty I can maintain their curriculum.
- As A Faculty I can approve or deny a student's course drop application.
- As A Faculty I can keep updated my students about upcoming curriculum activities like quizzes or important dates.
- As A Faculty I can keep track of a student's progress.

Student Use cases:

- As A Student I can attend classes without any hassle
- As A Student I can do my extracurricular activities.
- As A Student I can practice outdoor games.
- As A Student I can help my section as Class Representative
- As A Student I can seek for help anytime if I face problem.

Supervisor Use Cases:

- As A Supervisor I can control admin profile
- As A Supervisor I can add and remove courses
- As A Supervisor I can change any type or mistakes that has made during creation.

2.3.2 Non-Functional Requirements

The quality attributes that used for its functional behavior -

- **Performance**
- **Reliability**
- **Integrity**
- **Usability**
- **Maintainability**
- **Scalability**

Performance: What response times, processing capacity, or efficiency levels are expected?

As AURA uses offline-caching method to reduce server load, it will response faster than the existing server. Processing capacity will be minimal as everything will be processed within the user's device. Combining this two, AURA is much efficient than existing server.

Reliability: How will the system ensure stable and uninterrupted service?

As AURA uses offline caching method, it will ensure more stable and uninterrupted service.

Integrity/Security*: *What protections will safeguard data, authentication, authorization and privacy?

For storing data into database, we will use Cipher to encrypt the data and user will be authenticated using their credentials that provided by the facility.

Usability*: *What level of ease of use, accessibility and user experience should be maintained?

The dashboard and navigation must be understandable. User should use their respective functions without any issues. Icons and labels should be according to the features so user understands it. App color palette should be as like a user see. For example, CSE student and other users should see the dashboard blue while EEE student and other users should see the dashboard as orange (according to their department color). This app must maintain blue color tone which is AIUB official color palette. As this app uses offline-caching technology, when a user tries to login without internet connection should see the standby data that don't require active internet connection.

Maintainability**: **How will the system support future modifications, bug fixes, or upgrades?

As this project will follow scrum model, modification can be done via every sprint. Scrum ensures that the codebase remains clean and adaptable to future changes.

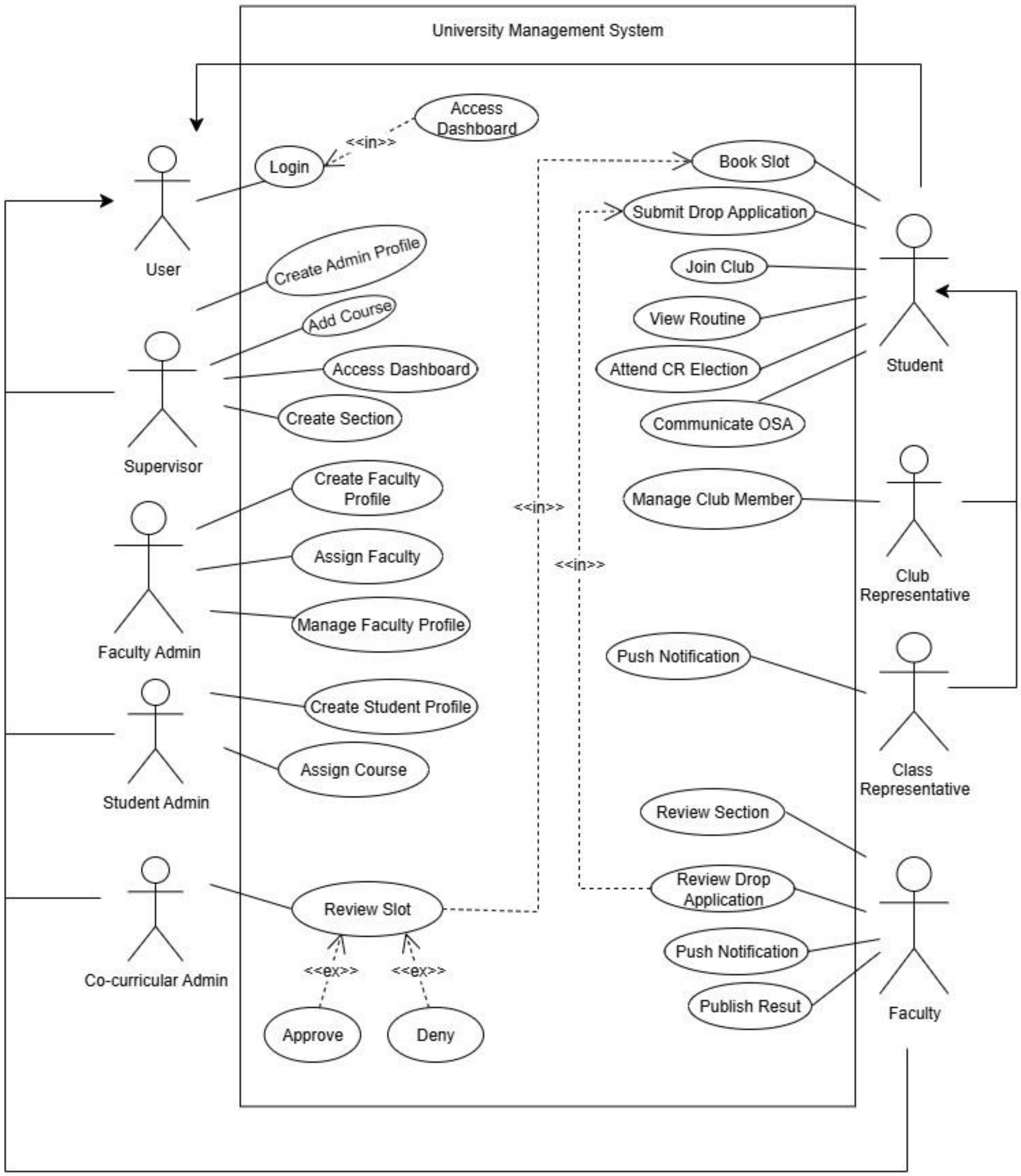
Scalability**: **How should the system grow to support more users, data, or extended features?

As this project will have offline-caching mechanism, that will reduce pressure on existing server and users can access stand-alone data without active internet connection. Also it will have a easy-go club joining feature, student can join clubs just scanning the QR-code from the booth. That will transfer general information automatically that includes student name, student ID and other information. Booking slot from the app is also an extended features.

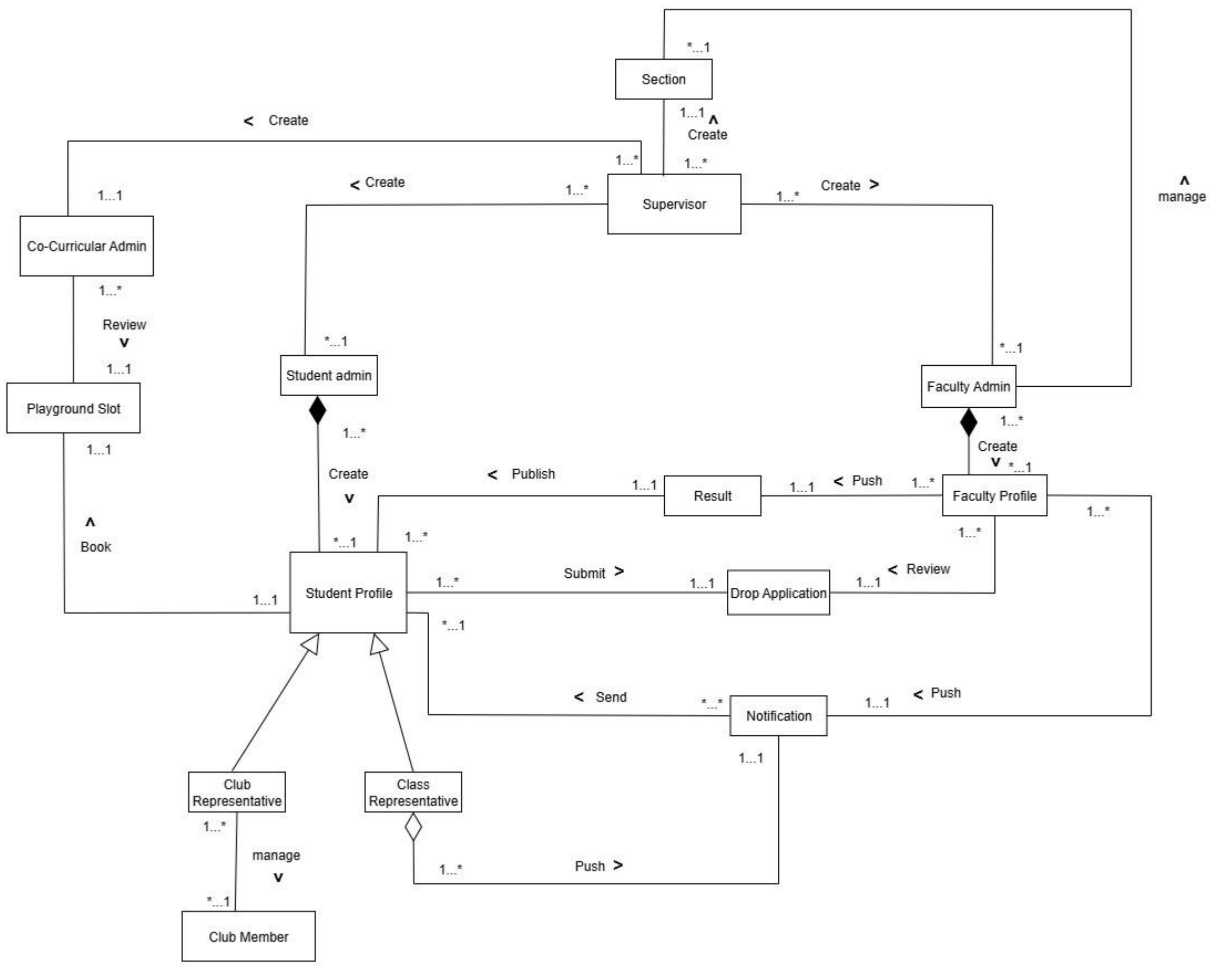
3. SOFTWARE DESIGN

3.1 System Design:

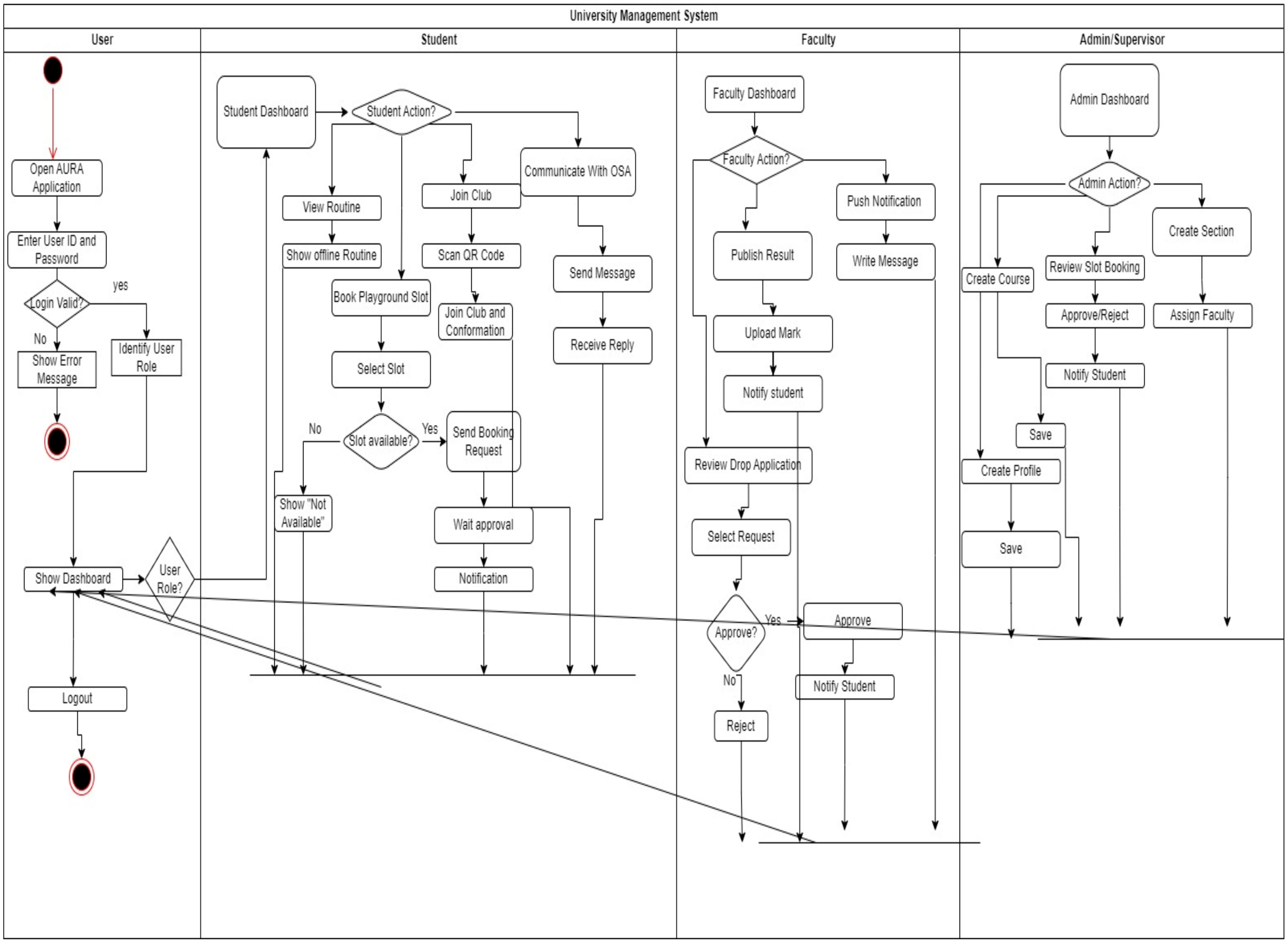
- **Use Case Diagram**



Class Diagram



Activity Diagram



3.2 UI / Wireframe Design using Figma

Template Name	AURA: Unified University Portal
Goals & Objective	Every university has it's own system to maintain. The goal of this project is to provide efficient portal access for individuals (student, faculties, supervisors) without pressurizing the main server. From this portal a user can - Login to portal which will be locally cached, only new data will be stored - Can push results [Faculty] - Can see results [Student] - Can book slot for playgrounds [Student] - Can push notifications [Faculty] [Class Representative] - Can maintain student or faculty profiles [Student Admin] [Faculty Admin] [Supervisor]
Product Type	Android APP
Domain	University Management System (UMS)
Access Layers	student, student admin, faculty, faculty admin, co-curricular admin, supervisor
Design Required	Yes

Phase	Item No.	Specification [Role]	User Story	Acceptance Criteria	Tags
Phase 1	1	Login UI [Student] [Faculty] [Student Admin] [Faculty Admin] [Co-Curricular Admin] [Supervisor]	- **As a** user, **I want to** access my dashboard, **so that** I can get access to my controls.	- Valid login credentials	loginSection

Phase 2	2	Dashboard [Student] [Student Admin] [Faculty Admin] [Supervisor]	- **As a** student, **I want to** access class routine, **so that** I can attend classes without any hassle. - **As a** student admin, **I want to** access my dashboard, **so that** I can create or modify student profiles. - **As a** faculty admin, **I want to** access my dashboard, **so that** I can create or modify faculty profiles.	- Login function check - If ID is student, load student functions - If ID is faculty, load faculty functions - If ID is faculty admin, load faculty admin functions	userDashboard
			-**As a** Supervisor, **I want to** access my dashboard, **so that** I can create admin profile and create or modify courses and sections.	- If ID is student admin, load student admin functions - If ID is cocurricular admin, load co-curricular admin functions	

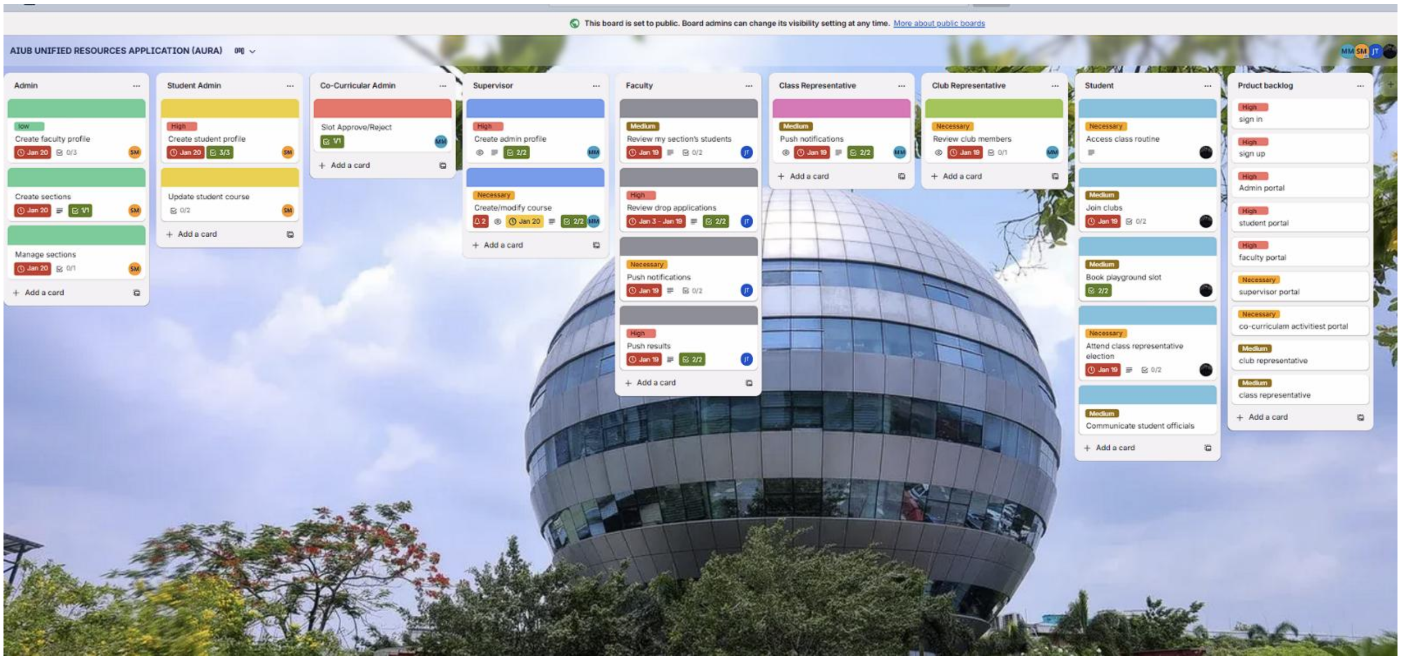
Phase 3	3	Create Profiles [Faculty Admin] [Student Admin] [Supervisor]	- **As a** faculty admin, **I want to** create a faculty profile, **so that** I can maintain faculty information. - **As a** student admin, **I want to** create a student profile, **so that** I can maintain them. - **As a** supervisor, **I want to** create new admin profiles, **so that** I can maintain the whole system.	- **Email** (input) field should only accept valid unregistered email address as input, otherwise will show an 'email already used' message. - **Password** (input) should be minimum 8 characters long. Otherwise show an invalid message. - **Re-password** (input) should match password. Otherwise show 'password do not match' message. - **User type** must be selected. Otherwise show 'Please select a user	createProfile

				type'. - **On clicking** signup button, credential will be stored in database, user will be redirected to his dashboard. - A copy of the login information will be sent to the registered email.	
	4	Add Course [Supervisor]	- **As a** supervisor, **I want to** add new courses to my system, **so that** I can give student admin access to assign the courses.	- Valid course name that should only accept **unregistered** course. - Name should have **at least** 5 characters long. - Every course must have a **course ID**. - Courses can have **prerequisites.** (Optional)	addCourse
	5	Create Sections [Supervisor]	- **As a** supervisor, **I want to** create sections, **so that** I can assign faculties.	- Valid course should be **selected** and **section name/character** should not exist for that specified course.	createSection

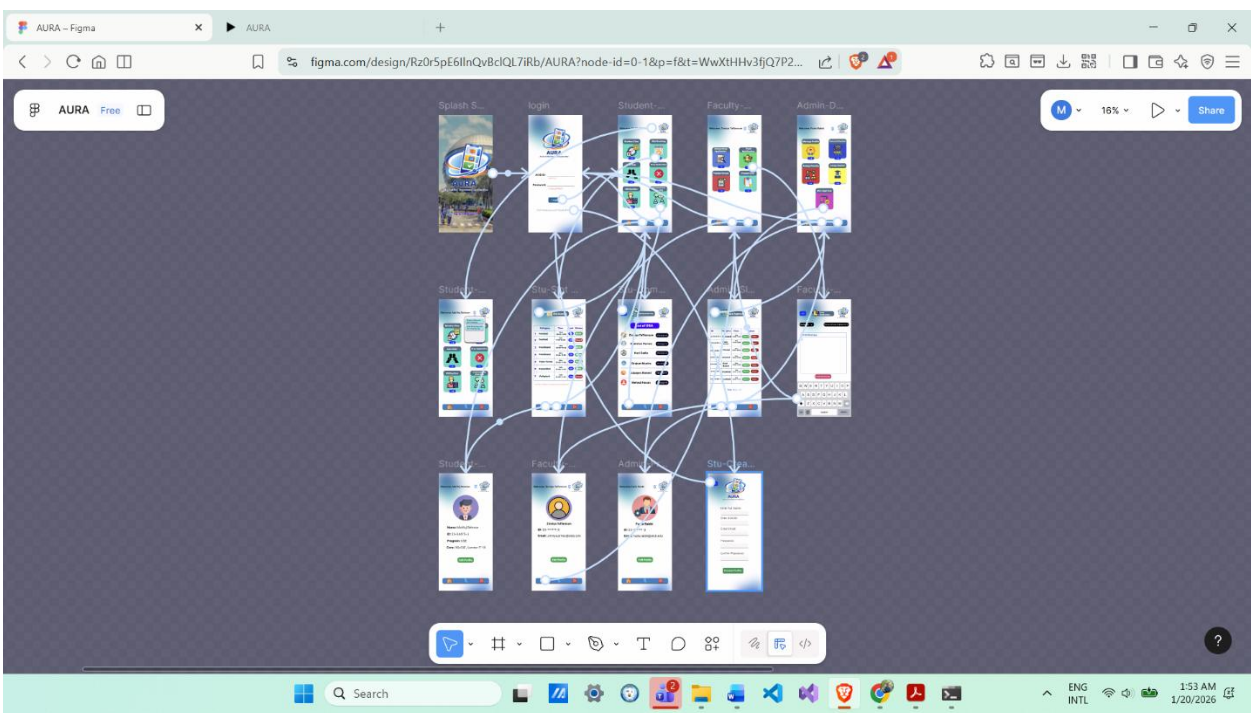
Phase 4	6	View Routine [Student]	- **As a** student, **I want to** view my class schedule easily, **so that** I can main my daily schedule.	- Valid student with **valid credentials**	userDashboard [Student]
	7	Submit Drop Application [Student]	- **As a** student, **I want to** submit a drop application to my faculty, **so that** I can withdraw from a course before the deadline.	- Valid **drop application**	
	8	Join Club [Student]	- **As a** student, **I want to** join club, **so that** I can main my extra curricular activities.	- Valid student - Valid club	
	9	Slot Booking [Student]	- **As a** student, **I want to** book slot for playground, **so that** I can play on the specified time.	- Valid Student - Valid slot type - Valid timeframe	
	10	Attend Class Representative Election [Student]	- **As a** student, **I want to** attend class representative election, **so that** I can be either candidate or cast my vote.	- Valid student - Valid section	
Phase 5	11	Review Section [Faculty]	- **As a** faculty, **I want to** review my sections, **so that** I can maintain their curriculum.	- Valid faculty - Valid section	userDashboard [Faculty]
	12	Review Drop Application [Faculty]	- **As a** faculty, **I want to** review my section's drop application, **so that** I can approve or deny a student's course drop application.	- Valid faculty - Valid **drop application** from valid student	

	13	Push notifications [Faculty]	- **As a** faculty, **I want to** push notifications to my section, **so that** I can keep updated my students about upcoming curriculum activities like quizzes or important dates.	- Valid faculty - Valid section - Valid notification	
	14	Publish results [Faculty]	- **As a** faculty, **I want to** publish results, **so that** I can keep track of a student's progress.	- Valid faculty - Valid course - Valid student	
Phase 6	15	Push notifications [Class Representative]	- **As a** class representative, **I want to** push notifications to my section on behalf of my faculty, **so that** I can help faculty to connect with students.	- Valid student - Valid class representative	userDashboard [Student]
	16	Review Club Members [Club Representative]	- **As a** club representative, **I want to** review my club members, **so that** I can manage club members who has joined already.	- Valid student - Valid club - Valid club representative	
	17	Communicate with OSA [Student]	- **As a** student, **I want to** communicate with OSA, **so that** I can seek for help anytime I face problem.	- Valid student - Valid message	
	18	Review Slot Booking [Co-Curricular Admin]	- **As a** co-curricular admin, **I want to** review slot booking, **so that** I can approve		

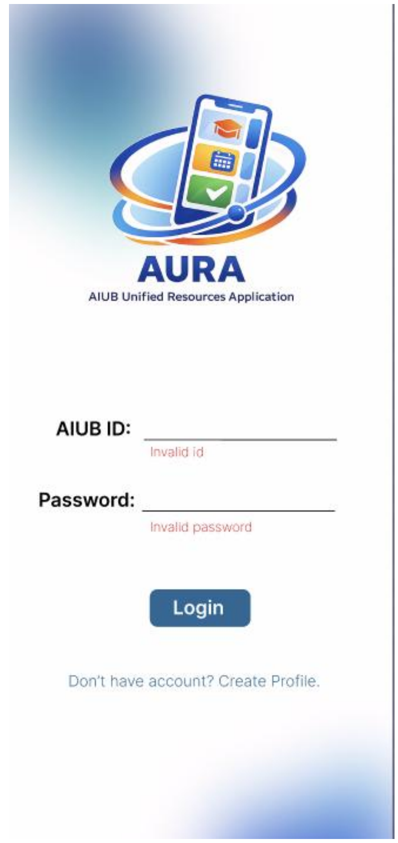
Trello Board

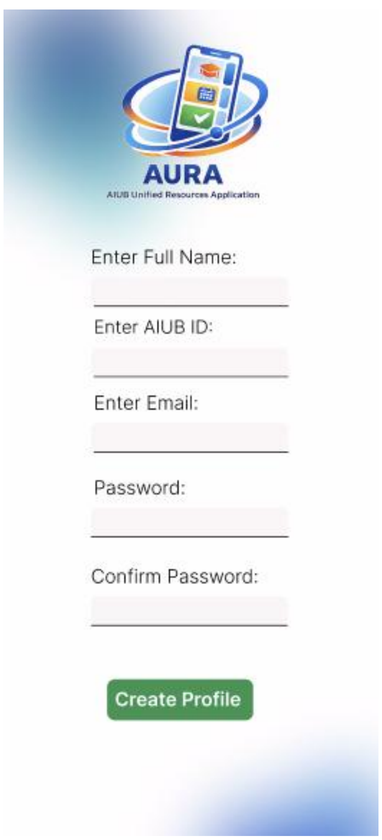


Figma Design

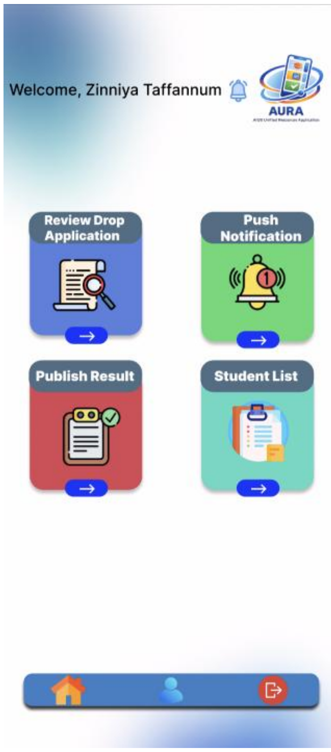


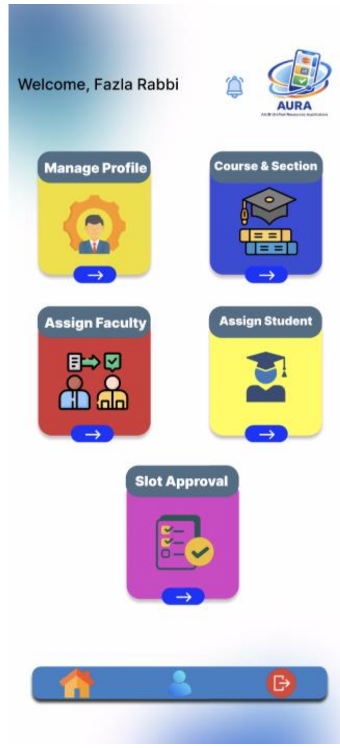

AURA
AIUB Unified Resources Application
Tap to Continue>>>


AURA
AIUB Unified Resources Application
AIUB ID:
Invalid id
Password:
Invalid password
Login
Don't have account? Create Profile.


AURA
AIUB Unified Resources Application
Enter Full Name:
Enter AIUB ID:
Enter Email:
Password:
Confirm Password:
Create Profile

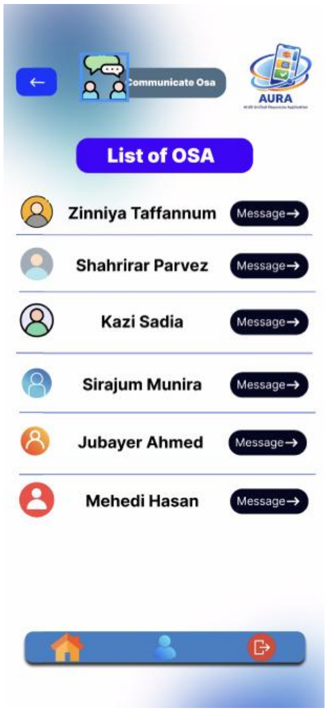

Welcome, Mahfuj Rahman
AURA
Routine View
Slot Booking
Join Club
Drop Application
CR Election
Communicate With OSA

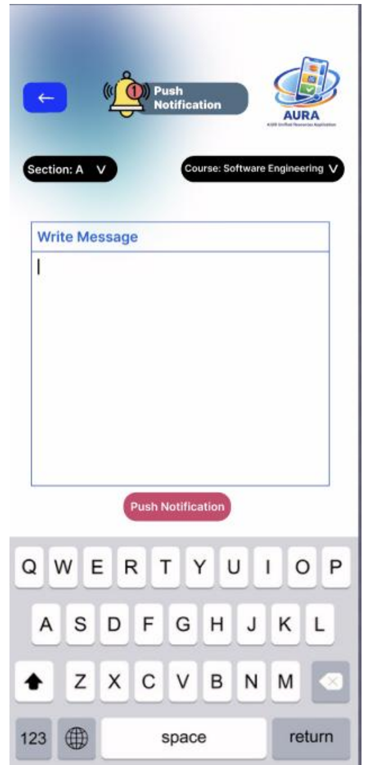

Welcome, Zinniya Taffannum
AURA
Review Drop Application
Push Notification
Publish Result
Student List


Welcome, Fazla Rabbi
AURA
Manage Profile
Course & Section
Assign Faculty
Assign Student
Slot Approval


Welcome, Mahfuj Rahman
Zinniya Taffannum send message
Fazla Rabbi Approve Your booking slot
Routine View
Join Club
Drop Application
CR Election
Communicate With OSA


Slot Booking
AURA
Category
Time
Limit
Status
1 Football Sun 10:00-11:00 2/5 Book
2 Football Sun 11:10-12:00 5/5 Closed
3 Pool Board Mon 10:00-10:30 8/15 Book
4 Pool Board Mon 10:30-11:00 2/5 Book
5 Table Tennis Mon 10:00-11:00 2/10 Book
6 BasketBall Sun 10:00-11:00 3/3 Book
7 Volleyball Sun 10:00-11:00 0/0 Closed
Terms & Conditions: Authority time change anytime


Communicate Osa
AURA
List of OSA
Zinniya Taffannum
Message
Shahrirar Parvez
Message
Kazi Sadia
Message
Sirajum Munira
Message
Jubayer Ahmed
Message
Mehedi Hasan
Message


Push Notification
AURA
Section: A
Course: Software Engineering
Write Message
Push Notification
Q W E R T Y U I O P
A S D F G H J K L
Z X C V B N M
123
space
return


Slot Approval
AURA
ID
Category
Time
Status
23-54475-3 Football Sun 10:00-11:00 Approve Reject
23-54479-3 Table Tennis Sun 11:10-12:00 Approve Reject
23-54389-3 Volleyball Mon 10:00-10:30 Approve Reject
23-54476-3 Basketball Mon 10:30-11:00 Approve Reject
23-53301-3 Pool Board Mon 10:00-11:00 Approve Reject
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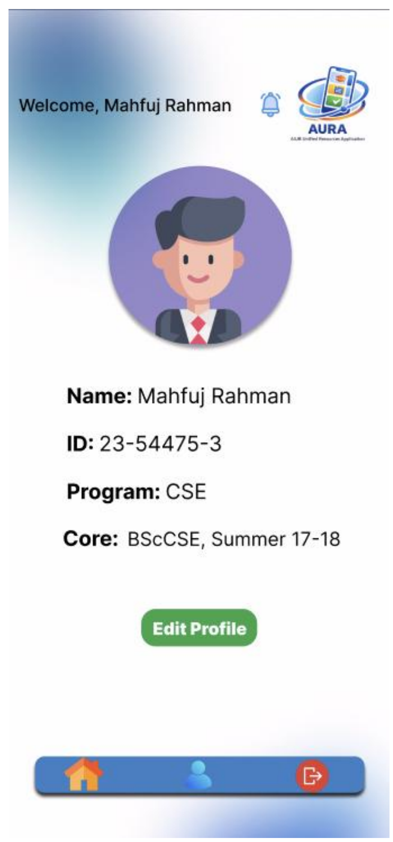
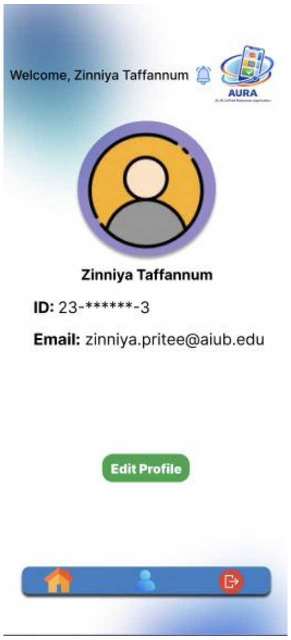
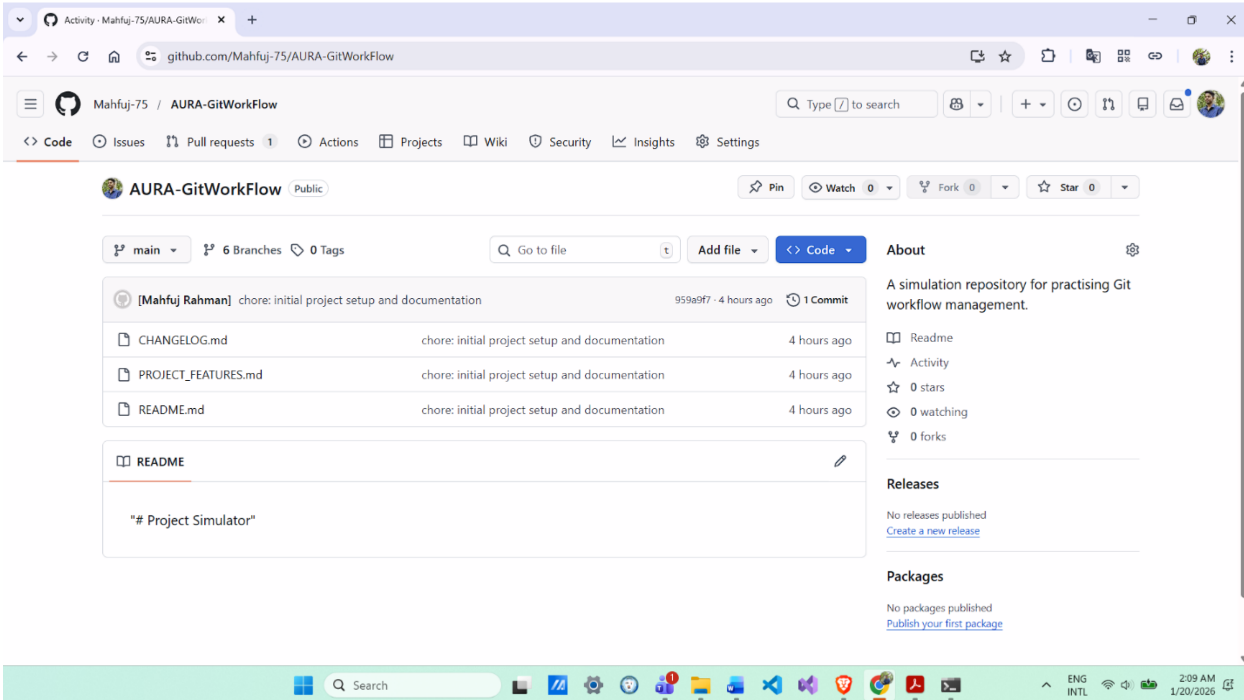
23-54459-3 Football Sun 10:00-11:00 Approve Reject
Page 1 2 3....

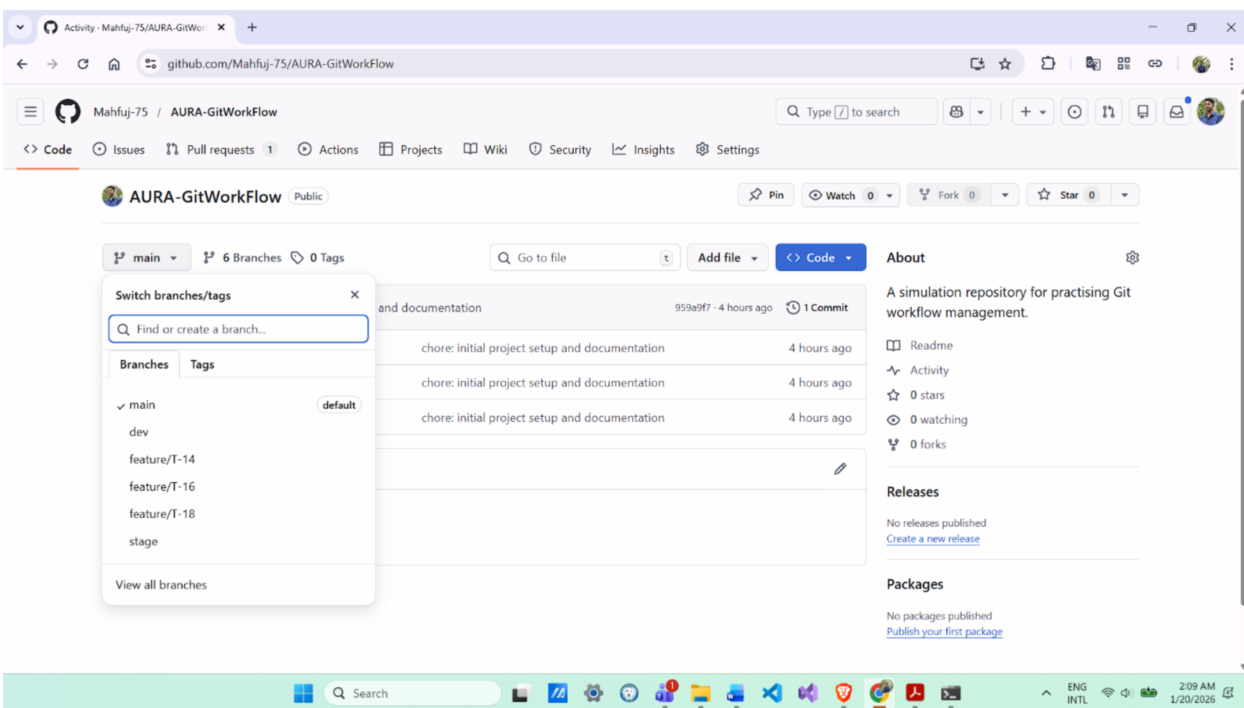

Welcome, Mahfuj Rahman
AURA
Name: Mahfuj Rahman
ID: 23-54475-3
Program: CSE
Core: BScCSE, Summer 17-18
Edit Profile

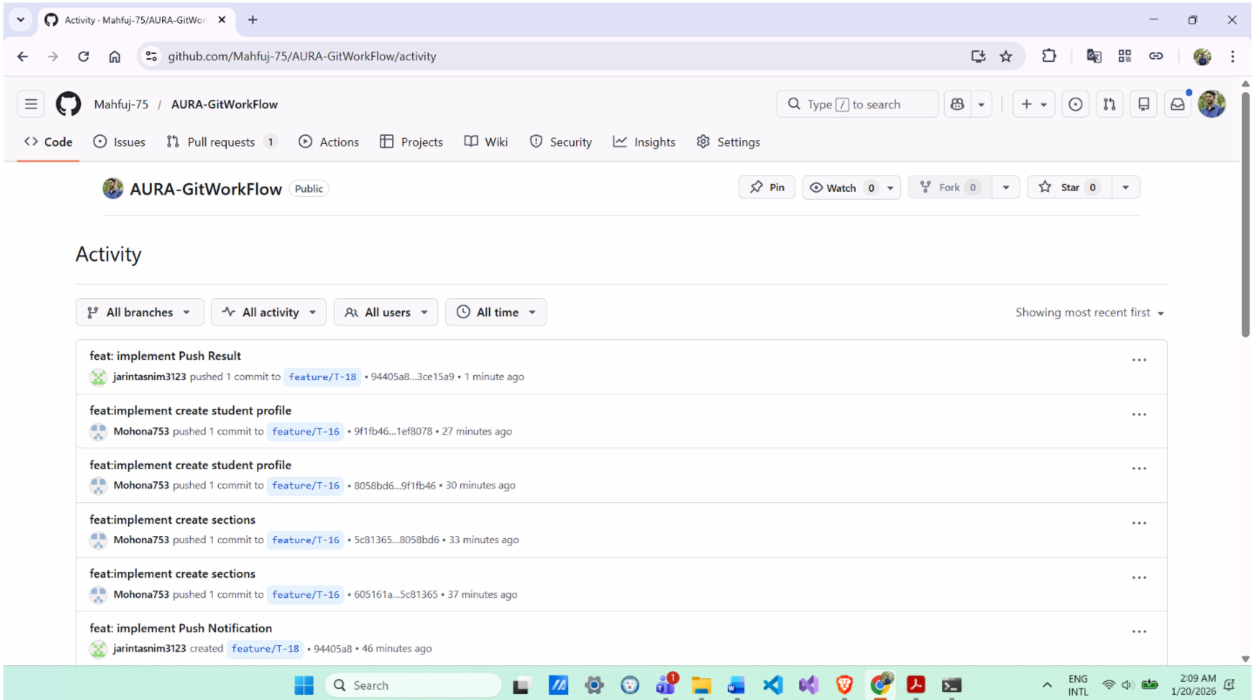

Welcome, Zinniya Taffannum
AURA
Zinniya Taffannum
ID: 23-******-3
Email: zinniya.pritee@aiub.edu
Edit Profile

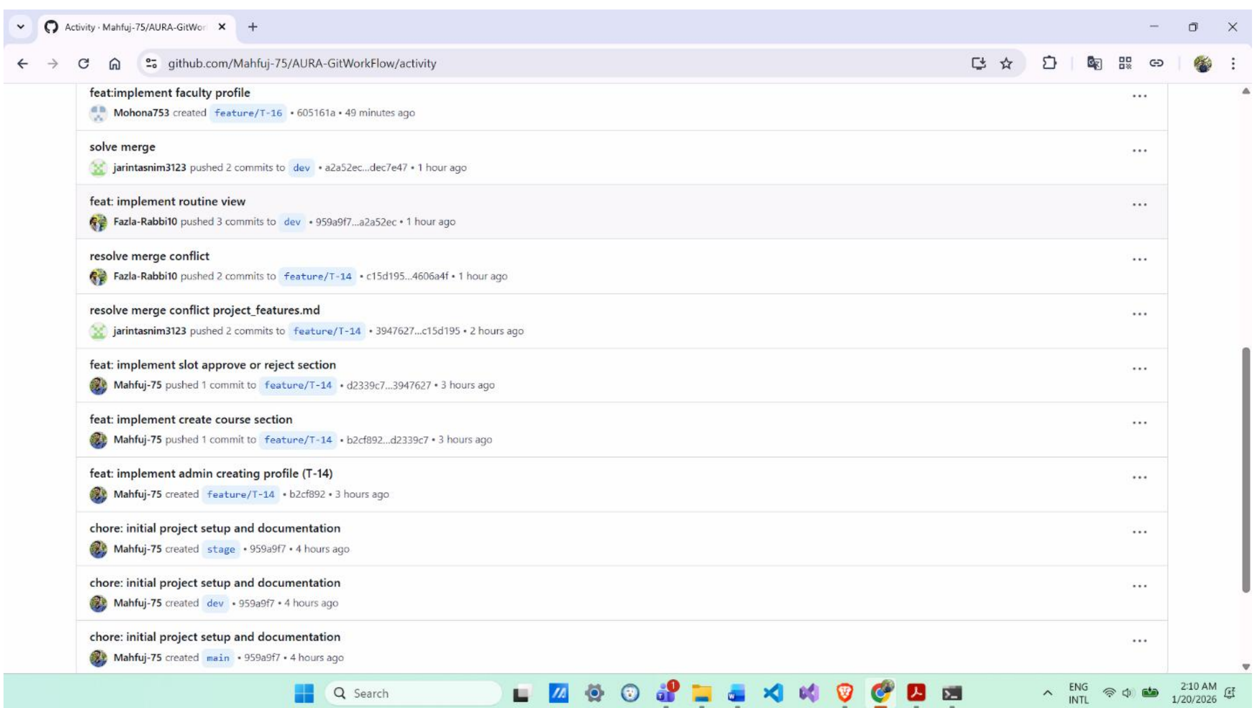

Welcome,Fazla Rabbi
AURA
Fazla Rabbi
ID: 23-******-3
Email: fazla.rabbi@aiub.edu
Edit Profile

4. GIT WORKFLOW






Activity · Mahfuj-75/AURA-GitWor
github.com/Mahfuj-75/AURA-GitWorkFlow/activity
Mahfuj-75 / AURA-GitWorkFlow
Type / to search
Code
Issues
Pull requests 1
Actions
Projects
Wiki
Security
Insights
Settings
AURA-GitWorkFlow
Public
Pin
Watch 0
Fork 0
Star 0
Activity
All branches
All activity
All users
All time
Showing most recent first
feat: implement Push Result
jarintasnim3123 pushed 1 commit to feature/T-18 • 94405a8...3ce15a9 • 1 minute ago
feat:implement create student profile
Mohona753 pushed 1 commit to feature/T-16 • 9f1fb46...1ef8078 • 27 minutes ago
feat:implement create student profile
Mohona753 pushed 1 commit to feature/T-16 • 8058bd6...9f1fb46 • 30 minutes ago
feat:implement create sections
Mohona753 pushed 1 commit to feature/T-16 • 5c81365...8058bd6 • 33 minutes ago
feat:implement create sections
Mohona753 pushed 1 commit to feature/T-16 • 605161a...5c81365 • 37 minutes ago
feat: implement Push Notification
jarintasnim3123 created feature/T-18 • 94405a8 • 46 minutes ago
Search
ENG
INTL
2:09 AM
1/20/2026


Activity · Mahfuj-75/AURA-GitWor
github.com/Mahfuj-75/AURA-GitWorkFlow/activity
feat:implement faculty profile
Mohona753 created feature/T-16 • 605161a • 49 minutes ago
solve merge
jarintasnim3123 pushed 2 commits to dev • a2a52ec...dec7e47 • 1 hour ago
feat: implement routine view
Fazla-Rabbi10 pushed 3 commits to dev • 959a9f7...a2a52ec • 1 hour ago
resolve merge conflict
Fazla-Rabbi10 pushed 2 commits to feature/T-14 • c15d195...4606a4f • 1 hour ago
resolve merge conflict project_features.md
jarintasnim3123 pushed 2 commits to feature/T-14 • 3947627...c15d195 • 2 hours ago
feat: implement slot approve or reject section
Mahfuj-75 pushed 1 commit to feature/T-14 • d2339c7...3947627 • 3 hours ago
feat: implement create course section
Mahfuj-75 pushed 1 commit to feature/T-14 • b2cf892...d2339c7 • 3 hours ago
feat: implement admin creating profile (T-14)
Mahfuj-75 created feature/T-14 • b2cf892 • 3 hours ago
chore: initial project setup and documentation
Mahfuj-75 created stage • 959a9f7 • 4 hours ago
chore: initial project setup and documentation
Mahfuj-75 created dev • 959a9f7 • 4 hours ago
chore: initial project setup and documentation
Mahfuj-75 created main • 959a9f7 • 4 hours ago
Search
ENG
INTL
2:10 AM
1/20/2026

5. SOFTWARE TESTING

<table>
<tr><td colspan="3">Project Name: AURA</td><td colspan="2">Test Designed by: Rabbi</td></tr>
<tr><td colspan="3">Test Case ID: TC_1</td><td colspan="2">Test Designed Date: 01/09/2025</td></tr>
<tr><td colspan="3">Test Priority (Low, Medium, High): Medium</td><td colspan="2">Test Executed By: Rabbi</td></tr>
<tr><td colspan="3">Module Name: Login Session</td><td colspan="2">Test Execution Date: 10/01/2026</td></tr>
<tr><td colspan="3">Test Title: Verify login with valid username and password</td><td colspan="2"></td></tr>
<tr><td colspan="3">Description: Test the app login page</td><td colspan="2"></td></tr>
<tr><td colspan="3">Precondition: The user has a valid user ID and password</td><td colspan="2"></td></tr>
<tr><td colspan="3">Dependencies: None</td><td colspan="2"></td></tr>
<tr><td>Test Steps</td><td>Test Data</td><td>Expected Results</td><td>Actual Results</td><td>Status (Pass/Fail)</td></tr>
<tr><td>1. Open the app
2. Enter User ID
3.Enter password
4. Click Sign In</td><td>Username: 23-53013-3
Password: 321@!#Rabbi</td><td>The user should log in to the application and see the expected dashboard</td><td>As expected</td><td>Pass</td></tr>
</table>

<table>
<tr><td colspan="3">Project Name: AURA</td><td colspan="2">Test Designed by: Rabbi</td></tr>
<tr><td colspan="3">Test Case ID: TC_2</td><td colspan="2">Test Designed Date: 02/09/2025</td></tr>
<tr><td colspan="3">Test Priority (Low, Medium, High): High</td><td colspan="2">Test Executed By: Rabbi</td></tr>
<tr><td colspan="3">Module Name: Dashboard Initiation</td><td colspan="2">Test Execution Date: 11/01/2026</td></tr>
<tr><td colspan="3">Test Title: Verify dashboard initialization</td><td colspan="2"></td></tr>
<tr><td colspan="3">Description: Test the dashboard functions loading properly</td><td colspan="2"></td></tr>
<tr><td colspan="3">Precondition: The user has a valid logged in session</td><td colspan="2"></td></tr>
<tr><td colspan="3">Dependencies: Login Page</td><td colspan="2"></td></tr>
<tr><td>Test Steps</td><td>Test Data</td><td>Expected Results</td><td>Actual Results</td><td>Status (Pass/Fail)</td></tr>
<tr><td>1. Open the app
2. Enter Faculty ID
3.Enter password
4.Click Sign In</td><td>Username: 2022-5259-2
Password: 321@!#Faculty</td><td>The user should see the available faculties function in their dashboard</td><td>Student functions loaded instead of faculty functions</td><td>Fail</td></tr>
</table>

Project Name: AURA			**Test Designed by:** Rabbi	
Test Case ID: TC_3			**Test Designed Date:** 03/09/2025	
Test Priority (Low, Medium, High): Medium			**Test Executed By:** Rabbi	
Module Name: Drop Review			**Test Execution Date:** 12/01/2026	
Test Title: Verify faculty can approve drop				
Description: Test the faction drop accept/reject functionality				
Precondition: The user has a valid logged in session as faculty				
Dependencies: Login Page				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Open the app 2. Login as Faculty 3. Navigate to the section 4. Click accept drop button	NONE	The drop application should be accepted and student drop application should be approved	Reject function triggered instead of approve	Fail

Project Name: AURA	**Test Designed by:** Rabbi
Test Case ID: TC_4	**Test Designed Date:** 02/09/2025
Test Priority (Low, Medium, High): Low	**Test Executed By:** Rabbi
Module Name: Push notifications (Faculty view)	**Test Execution Date:** 11/01/2026
Test Title: Verify notification system	
Description: Test the notification system from faculty dashboard	
Precondition: The user has a valid logged in session as faculty	
Dependencies: Login Page	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open the app 2. Login as faculty 3. Navigate to the section 4. Press Push Notification button 5. Enter notification message	Notification Message: Hello this is a test push	The user should see successful message and add the notification data into the database	As expected	Pass

Project Name: AURA			**Test Designed by:** Rabbi	
Test Case ID: TC_5			**Test Designed Date:** 01/09/2025	
Test Priority (Low, Medium, High): Medium			**Test Executed By:** Rabbi	
Module Name: Publish results [Faculty]			**Test Execution Date:** 14/01/2026	
Test Title: Verify publishing results				
Description: Test publishing results				
Precondition: The user has a valid user ID.				
Dependencies: None				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1.Open the app 2.Enter User ID 3.Navigate section 4. Publish results	Username: 23-53047-3 Password: 123chodu	The user should be able to upload the results	As expected	Pass

Project Name: AURA			**Test Designed by:** Mahfuj	
Test Case ID: TC_6			**Test Designed Date:** 02/09/2025	
Test Priority (Low, Medium, High): High			**Test Executed By:** Mahfuj	
Module Name: Push notifications (Representative view)			**Test Execution Date:** 14/01/2026	
Test Title: Push notifications [Class Representative]				
Description: Test the notification system from student dashboard				
Precondition: The user has a valid logged in session				
Dependencies: Login Page				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Open the app 2.Log in with ID 3. Navigate to the section 4. Press Push Notification button 5. Enter notification message	Notification Message: Hello this is a test push	The user should see successful message and add the notification data into the database	As expected	Pass

Project Name: AURA	**Test Designed by:** Mahfuj
Test Case ID: TC_7	**Test Designed Date:** 03/09/2025
Test Priority (Low, Medium, High): Medium	**Test Executed By:** Mahfuj
Module Name: Review Club Members [Club Representative]	**Test Execution Date:** 12/01/2026
Test Title: Verify that Club representatives can review club members	
Description: Test the club representatives review system	
Precondition: The reviewer has a valid logged in session as a student	
Dependencies: Login Page	

Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Open the app 2. Login as student 3. Navigate to the section 4.Click review button	NONE	The review button should take the user to a text box where the user can upload reviews	able to upload opinions about about club members	Pass

Project Name: AURA	**Test Designed by:** Mahfuj
Test Case ID: TC_8	**Test Designed Date:** 02/09/2025
Test Priority (Low, Medium, High): Low	**Test Executed By:** Mahfuj
Module Name: Communicate with OSA [Student]	**Test Execution Date:** 14/01/2026
Test Title: Verify student can communicate with OSA	
Description: Test that OSA replies within an hour	
Precondition: The user has a valid logged in session as Student	
Dependencies: Login Page	

Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. Open the app 2. Login as Student 3. Navigate to the section 4. Message OSA	Message: Hello this is a test push	The User should see OSA should reply within an hour	As expected	Pass

Project Name: AURA	**Test Designed by**: Mahfuj
Test Case ID: TC_9	**Test Designed Date:** 07/09/2025
Test Priority (Low, Medium, High): Medium	**Test Executed By:** Mahfuj
Module Name: Submit Drop Application [Student]	**Test Execution Date:** 13/01/2026
Test Title: Verify valid drop application	
Description: Test the drop application function work properly	
Precondition: The user has a valid user ID and password	
Dependencies: None	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open the app 2. Login as Student 3. Click on the drop button 4. Processed to the drop application	username:23-53447-3 password: as@gh/23	The can submit a drop application, **so that** they can withdraw from a course before the deadline.	The student succeed to submit a drop application	pass

Project Name: **AURA**			Test Designed by: **Mahfuj**	
Test Case ID: TC_10			**Test Designed Date:** 08/09/2025	
Test Priority (Low, Medium, High): Medium			**Test Executed By:** Mahfuj	
Module Name: Join Club [Student]			**Test Execution Date:** 14/01/2026	
Test Title: Verify successful club joining with valid credentials				
Description: Test that a student can successfully send request to join an available club.				
Precondition: The user is logged in and the club they wish to join is active and accepting members.				
Dependencies: User must have a valid student account.				
Test Steps	**Test Data**	**Expected Result**	**Actual Result**	**Status (pass/fail)**
1. Open the app 2. Login as a Student 3. Select a club to join 4. Click the 'Join' button	username:23-54478-3 password: St#456$US	The application home page loads successfully. User is successfully authenticated and directed to the student dashboard. The system navigates to the selected club's details page, showing a "Join" button. The system processes the request and displays a success message.	The user successfully navigated the flow and received a confirmation message after joining the club.	pass

Project Name: AURA			**Test Designed by: Mohona**	
Test Case ID: TC_11			**Test Designed Date:** 09/09/2025	
Test Priority (Low, Medium, High): Medium			**Test Executed By:** Mohona	
Module Name: Playground Booking[Student]			**Test Execution Date:**15/01/2026	
Test Title: Verify playground slot booking				
Description: Test the playground booking function work properly				
Precondition: The user has a valid logged in session				
Dependencies: Valid playground slot available				
Test Steps	**Test Data**	**Expected Result**	**Actual Result**	**Status [pass/fail]**
1. Open the app 2. Login as Student 3.Click Playground Booking 4.Select slot and time then Click Book Now	username: 23-5349-3 password: as@gh/23	The student can book slot for playground, so that they can play on the specified time	System showed "Slot not available" and booking failed	fail

Project Name: AURA	**Test Designed by: Mohona**
Test Case ID: TC_12	**Test Designed Date:** 10/09/2025
Test Priority (Low, Medium, High): Medium	**Test Executed By:** Mohona
Module Name: Attend Class Representative Election [Student]	**Test Execution Date:** 16/01/2025
Test Title: Verify class representative election attendance	
Description: Test the election attendance function work properly	
Precondition: The user has a valid logged in session	
Dependencies: Valid election ongoing in student's section	

Test Steps	Test Data	Expected Result	Actual Result	Status [pass/fail]
1. Open the app 2. Login as Student 3. Click Election section 4. Check for ongoing election	username: 23-50001-3 password: as@jh56$	The student can attend class representative election, so that they can be either candidate or cast their vote	Student was not able to cast their vote in an on-going election.	fail

Project Name: AURA			**Test Designed by: Jarin**	
Test Case ID: TC_13			**Test Designed Date:** 05/09/2025	
Test Priority (Low, Medium, High): High			**Test Executed By:** Jarin	
Module Name: Add Course			**Test Execution Date:** 15/01/2026	
Test Title: Verify supervisor can add a new course				
Description: Test the functionality of adding a new course				
Precondition: The user has a valid logged in session as Supervisor				
Dependencies: Supervisor Dashboard				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. **Open the app** 2. **Login as Supervisor** 3. **Navigate to Dashboard** 4. **Click on “Add Course” option**	Course Name : Software Engineering	Course should be added successfully	As expected	Pass

Project Name: AURA			**Test Designed by: Jarin**	
Test Case ID: TC_14			**Test Designed Date:** 06/09/2025	
Test Priority (Low, Medium, High): Medium			**Test Executed By:** Jarin	
Module Name: Create Section			**Test Execution Date:** 11/09/2025	
Test Title: Verify faculty admin can create section for a course				
Description: Test the section creation functionality				
Precondition: The user has a valid logged in session as Supervisor				
Dependencies: Supervisor Dashboard				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1.Open the app Login as Supervisor 2.Navigate to Dashboard 3. **Enter valid course and new section name, then click Create Section**	Course : Software Engineering Section : P	The section should be created successfully and appear under the specified course.	"Invalid section name" appeared instead of creating new section	Fail

Project Name: AURA	**Test Designed by: Jarin**
Test Case ID: TC_15	**Test Designed Date:** 07/09/2025
Test Priority (Low, Medium, High): Low	**Test Executed By:** Jarin
Module Name: View Routine	**Test Execution Date:** 15/01/2026
Test Title: Verify student can view class routine	
Description: Test the routine viewing functionality	
Precondition: The user has a valid logged in session as a Student	

Dependencies: Student Dashboard				
Test Steps	**Test Data**	**Expected Results**	**Actual Results**	**Status (Pass/Fail)**
1. **Open the app** 2. **Login as Student** 3. **Navigate to Dashboard** 4. **Click on "View Routine" option**	NONE	The student's class routine should be displayed properly according to their enrolled courses.	"No routine available" showed	Fail

6. CONCLUSION

The AURA project is designed to make university life easier for everyone. It is a unified platform for students, faculties and admins to manage academic, administrative and co-curricular activities easily. Students can check their class routines, book playground slots, join clubs using QR code and see results easily. Faculties can publish results, send notices and review drop applications without any delay.

This system helps to solve many real problems faced by AIUB users such as server overload, difficulties in checking routines and the need to be present physically to book playground slots. By reducing server load and storing data, the app saves time and gives faster access to everyone. For this project we have selected scrum method, which allowed the team to work in small steps. This made the work more organized, flexible and easier to manage. Each sprint focused on a specific feature and daily meetings will help the team to improve the project.

Future Work

We aim to make AURA application more useful to make it more better for students, faculties and admins. Future work includes –

- **Online Payment System:** Students can pay fees or fines directly through the app also they can hold credits under their user profile for future usage (Buying foods from canteen).
- **Chat System:** Students can communicate with each others, even with faculties and OSA members.
- **Canteen Management System:** Students can purchase canteen foods from their application or they can schedule when they are planning to eat their food as per their class schedule.
- **Digitalized Lost & Found:** If anyone lost anything inside university campus, they can search for Lost & Found section and collect them physically with valid evidence.
- **Semester Calendar:** Where full semester plan will be available including university planned mid term exam and holidays.

We believe, these future works will make AURA more better and useful for our target users.